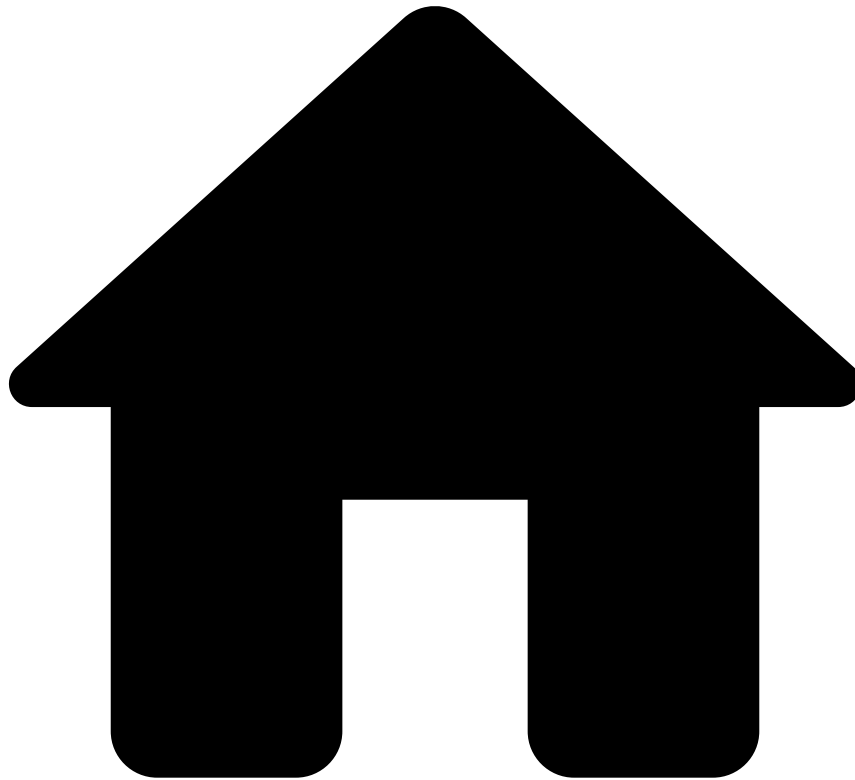


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- Building Rules

On this page

Building Rules

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[Models](#) are one way to contribute towards the score of a given decision, *fraud/not fraud* in this case, while the other are rules.

Rules add an extra layer of security on top of machine learning models by letting you define conditions that, when met, help you classify your events.

This tutorial stage will help you define and evaluate simple rules. However, rules in Pulse are a very robust and complex feature that allows you to do much more. Don't forget to check the [Rules](#) page for further details.

Defining rules

To create rules in Pulse, execute the following actions:

1. Create a [list](#) for suspicious merchants as follows:
 1. Name the list as *MerchantsAlert*. Leave all other fields with the default values.
 2. **Save and add items** with the following values: *Royal Fitness*, *WALGREENS* and *value_5*.
2. In the **Data Science** tab, under the Rules Projects section, add a new rules project, and configure it as follows:
 1. Name it *Rules*.
 2. Copy the schema from *TransactionsFeatures*.
 3. Set the **Target** to the *Fraud* field.
 4. Add an **Action** named *MerchantsAlert*.
3. Add a new rule named *MerchantsAlert* to the rules project and define its behavior as follows:
 1. In the **Variables** field, set an *isSuspicious* variable that reflects if the *MerchantName* of the transaction is contained in the *MerchantsAlert* list.
 2. In the **Condition** field, set the rule to trigger if the merchant is suspicious and the *TransactionAmount* is greater than 50.
 3. In the **Score Contribution** field, set the contribution to the *Fraud* score as 200.
 4. In the **Actions** field, select the *MerchantsAlert* action. This means that the rule will trigger this action if the Condition is met.

This rule is not very smart and serves as only as a demonstration of how you can configure a rule. This rule's behaviour defines that whenever a transaction with an amount above 50 was performed at a suspicious merchant, then Pulse should trigger an action and add 200 to the event's score. However, the rule does not mark the transaction as *Fraud* since it this only a strong indication of fraud and not a certainty.
4. Add a new rule to the rules project. This rule is simpler than the first one, but it actually marks transactions as *Fraud* based on the observation that fraud is more prevalent in transactions with a higher amount. Configure it as follows:
 1. Name it *TransactionAmount*.
 2. In the **Condition** field, trigger the rule when the *TransactionAmount* is greater than 600.
 3. Trigger the **Action** named *Block*.
 4. Mark the transaction as *Fraud*.

Exporting and evaluating rules

Now that you have defined a block of rules, you must create a [snapshot](#) so that you can use them in a workflow. While you are configuring the snapshot, you can also evaluate the rules performance over a specific data source as follows:

1. Publish your application.
2. In the *Rules* rules project, add a new snapshot called *Snapshot*.
 1. Include all your rules in the snapshot.
 2. **Evaluate** it using the *TransactionsFeatures* data source.
3. Once the evaluation finishes, open the snapshot and notice that there are 3 kinds of evaluations:
 1. **Block Evaluation**: shows the evaluation regarding the target field (*Fraud*) after all rules are executed.

2. **Block Rule Evaluation:** similar to block evaluation, but instead of seeing all rules as a black box, you can drill down into the performance of each rule.
3. **Individual Rule Evaluation:** similar to block rule evaluation, however the rules are not evaluated in sequence, but rather in isolation. This means that the result of a rule does have an impact on the evaluation of the next one.

In this scenario, you won't see any difference between individual rule evaluation and block rule evaluation because the first rule being applied does not mark events as *fraud* or *not fraud*.